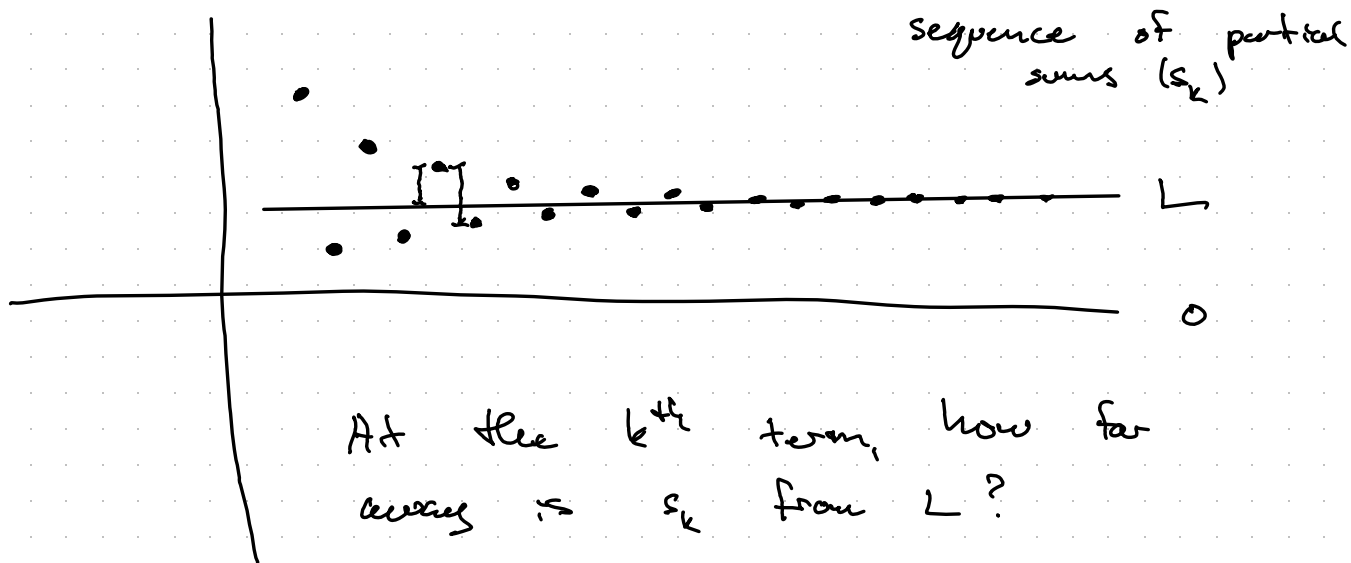


Alternating series error bound - S.S



Asking how large $|s_k - L|$ is.

Observe that $|s_k - L| \leq |s_{k+1} - s_k| = |a_{k+1}|$

Error bound $|s_k - L| \leq |a_{k+1}|$

Example: Bound the error of the estimation

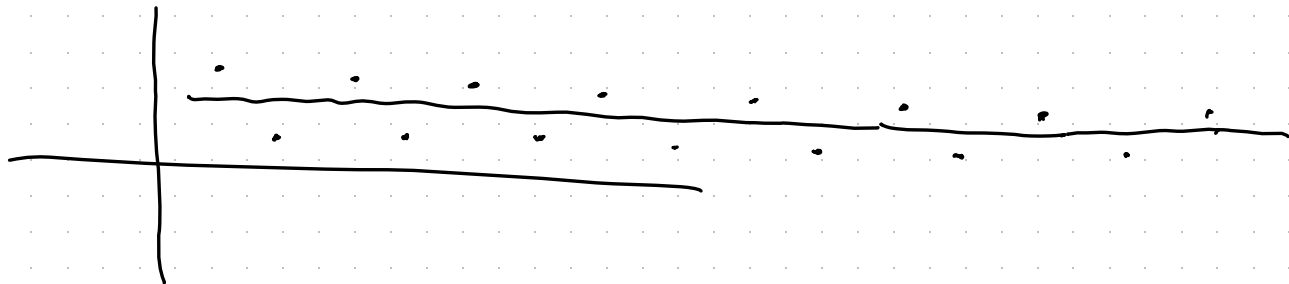
of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ by s_{10} .

$$\left| s_{10} - \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \right| \leq |a_{11}| = \left| \frac{(-1)^{12}}{11^2} \right| = \frac{1}{11^2} = \frac{1}{121}$$

This says the value of the complete

sum $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ is within $\frac{1}{121}$ of s_{10} .

Example: $\sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln(n)}$ converges by the AST



How many terms must we add before we are confident that the result is within $\frac{1}{10}$ of the actual value of the series?

$$\rightarrow \text{know } \left| s_k - \sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln(n)} \right| \leq |a_{k+1}| = \frac{1}{\ln(k+1)} \stackrel{?}{<} \frac{1}{10}$$

Need $10 < \ln(k+1)$, so $e^{10} < k+1$. Thus need k at least $e^{10} - 1 \approx 22025.466 \dots$, take $k = 22026$.